

Software Requirements for Memory Flash

Memory flash

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Overview

This outlines the requirements for the software implementation of a game called Memory flash. The game challenges players to remember and repeat an increasing sequence of colors. Each round, the sequence grows, testing the player's memory and attention.

The game

A player is presented with a sequence of colored buttons that light up in a pattern, generated by the game. The player must repeat the pattern by clicking the buttons in the exact order. Each round the sequence increases by one color; if the player reproduces the sequence correctly, they advance to the next round, but if they make a mistake, the game ends.

Software implementation

UI elements

- 3x3 Grid
 - 5 buttons in total
 - The 4 cornered ones are for the random colors
 - The middle button is the start button
 - Input validation
- Row of buttons on top of screen that contain
 - Message display area (instructions, game over notification)
 - Score display
 - Visual cues to show invalid input or mistakes

Game Rules

- The game begins only once the player clicks start, before this the game is inactive.
- Each round, the game generates a sequence of colors by lighting up buttons in order.
- The player must then repeat the sequence by clicking the corresponding colored buttons in the exact order.
- If the player repeats the sequence correctly, the game advances to the next round by adding one additional color to the sequence.
- If the player clicks the wrong color, or out of order, the game ends with a "Game Over" message, and visual feedback indicating the mistake.